→ Planning : in AI is about decision making tasks performed by robots or comp. to achieve a specific goal.

↓

execution = seq. of actions ⇒ ↑ likelihood of task completion.

Classical Planning Environment = full observable, deterministic, finite, static & discrete

others = non-classical

↳ diff set of algos & agent designs

→ Real world problems are ? nearly decomposable

↓

large complex problems ⇒ into smaller subproblems

? → subgoals are solved independently by planner
⇒ largely independent but may interfere

→ Planning systems need a way to describe problems
⇒ Expressive but Restrictive
↙ describe wide variety of probs.
↳ to allow algos to operate over it

STRIPS : STanford Research Inst Problem Solver

↑ basic rep. lang for classic planners.

Components of STRIPS

- Set of States
  = conj of positive ground literals
  → describes current world using facts = True
  At (Home) ∧ Have (Banana)
- Goals
  = partially specified state rep as conj of literals
  * can have variables
  → what must be true in final state
  At (Home) ∧ ~ Have (Banana)
  At (x) ∧ Sell (x, Banana)
- Actions
  - Precondn
    ↙ condn. that need to be True before Action
  - Effect
    ↙ how world changes after action

Action : Buy (x)
Precondition: At (p), Sell (p, x)
Effect: Have (x)

Challenges of AI & Planning:

1. Closed world Assumption - assumes that the world model contains everything that the robot needs to know; no surprise

2. Framing Problem: how to represent a real world situation in a way that is computationally tracable (Expressive + Restrictive)

Planning with State Space Search:

Forward (Progression)

Backward (Regression)

Initial State

Actions

Goal Test

Step Cost

Planning

TOP — Total Order

POP — Partial Order

Total Order Planning : only explore linear seq. of action from start → goal.

⇒ Cannot take advantage of problem decomposition

Partial Order Planning : works on prob decomposition

⇒ ie divide problem into parts & achieve the subgoals independently

⇒ solve subproblems with subplans ⇒ combine + reorder on basis of goal req.

⇒ does not specify which action will come out first of the 2 that are placed.

POP

Planning — State
— goal
— Action — Pre
— Eff

Eg: Putting on Pair of Shoes

Goal: RightShoeOn $\wedge$ LeftShoeOn

Action:

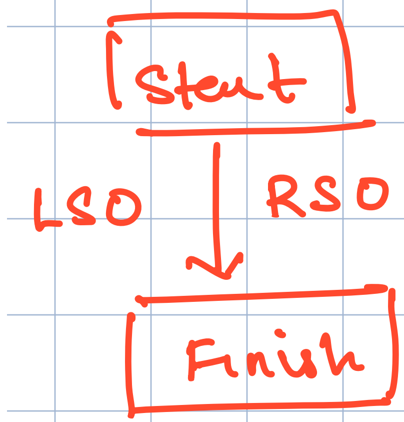
1. RightShoe
   Precondition: RightSockOn
   Effect: RightShoeOn

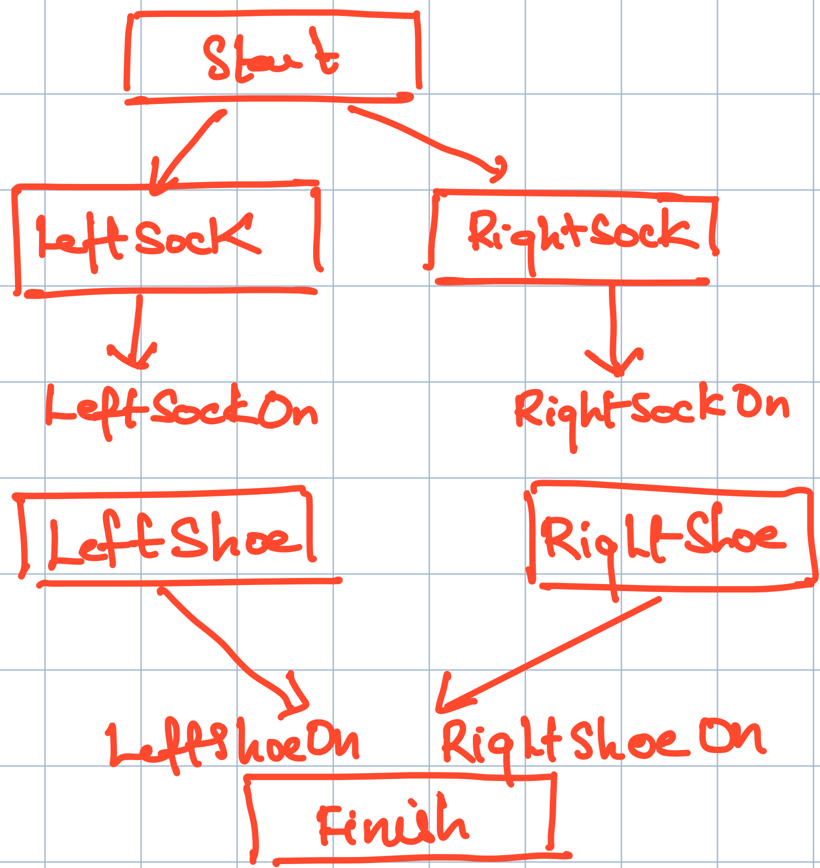
2. LeftShoe
   Precondition: LeftSockOn
   Effect: LeftShoeOn

3. RightSock
   Precondition: None
   Effect: RightSockOn

4. LeftSock
   Precondition: None
   Effect: LeftSockOn

POP




TOP (6 diff. plans)


| | | | | |
|---|---|---|---|---|
| RSock | L So | RSo | LSo | RSo |
| LSock | RSo | LSo | LSh | RSh |
| LShoe | RSh | RSh | RSo | LSo |

$\boxed{\text{Right Shoe}}$

↓

$\boxed{\text{Finish}}$

RShoe LSh LSh RSh LSh

* POP & Hirearchial look similar but

⇒ POP = breaking goal → subgoals

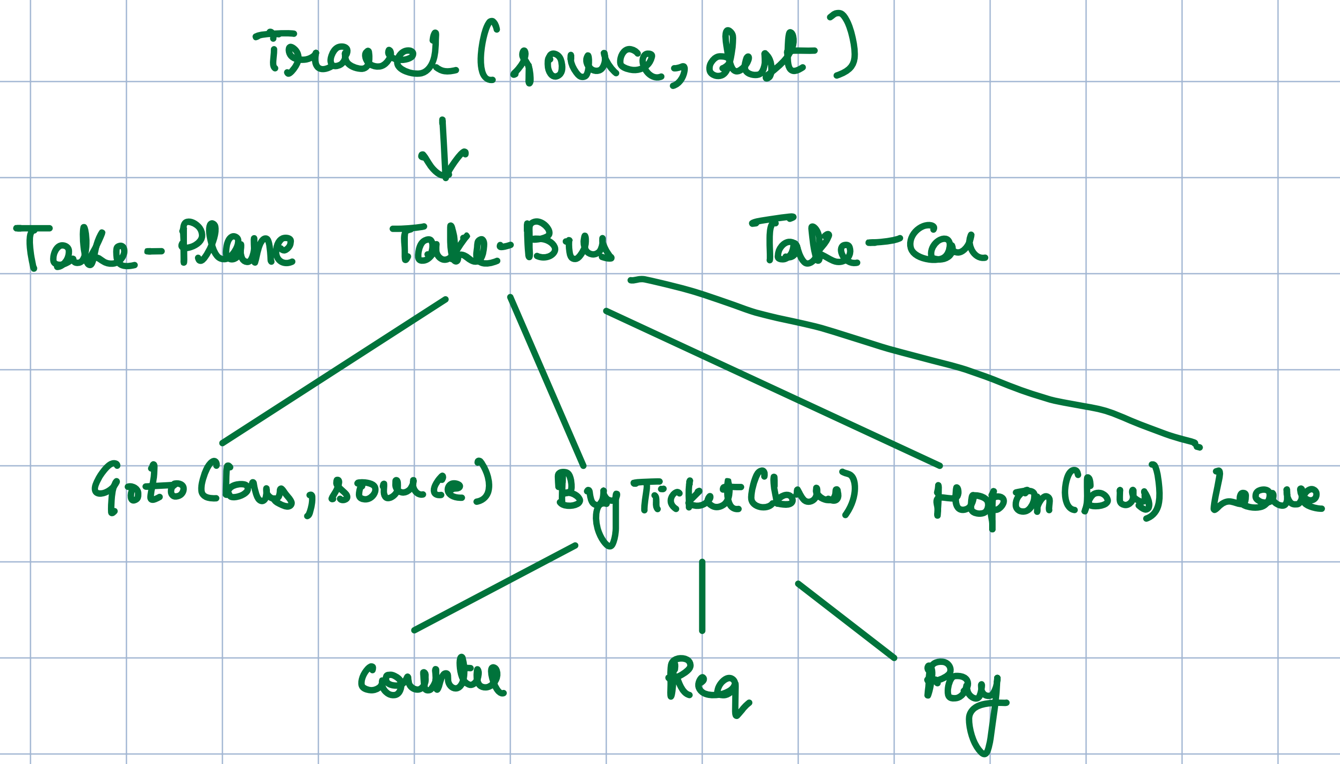
Shoes On
- RSO
- LSO

⇒ Hirearchial = abstract → lower level

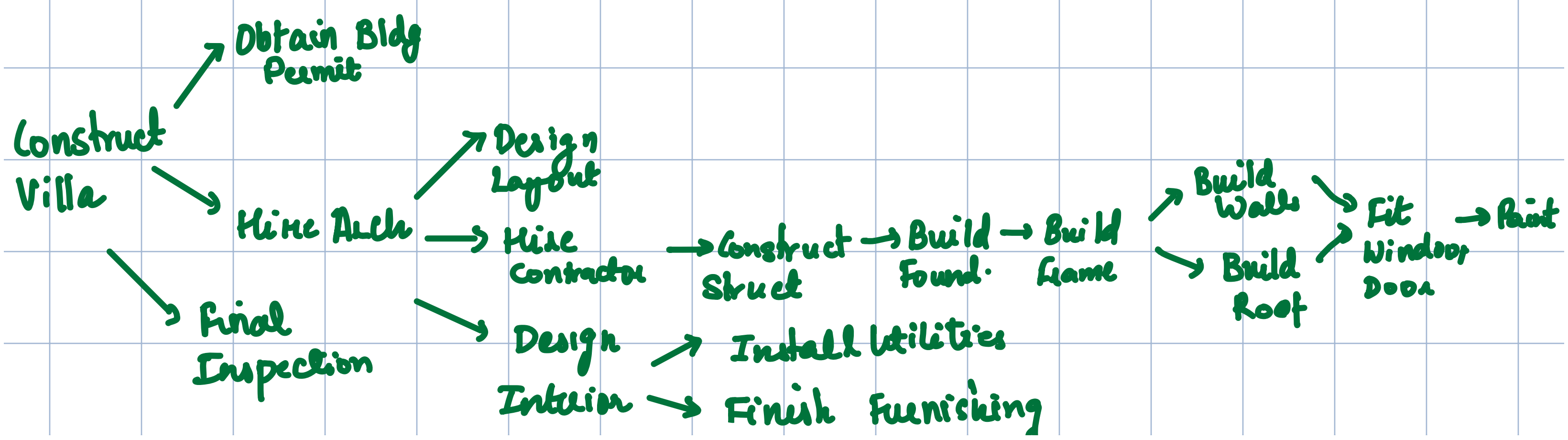
Travel
- T by Bus
- T by Train
- T by Car

# Hierarchial Planning

⇒ Plans are organized in hierarchial format
⇒ works on decomposition




eg: Construct villa after buying land



# Conditional Planning

↳ work regardless of outcome of the action → as contingency branches

↳ deals with uncertainty by determining what happens in env. at predetermined points in plan

↳ fully observable + non-deterministic → same action → diff outcomes

↳ I/P action ⇒ should handle every outcome

Classic fails → Action fail
→ uncertainties in env
→ outcome not as predicted

# Contingent Planning

↳ work under partial observability with sensing agents, that sense meaningful facts

* Conditional = uncertain conseq. (branches created by outcomes)
  Contingent = uncertain conditions (branches created by condn.)

# STRIPS v/s ADL

| | SR. NO. | PARAMETERS | STRIPS | ADL |
|---|---|---|---|---|
| 1 | 1 | Stands for | **ST**anford **R**esearch **I**nstitute **P**roblem **S**olver | **A**ction **D**escription **L**anguage |
| 2 | 2 | Literals allowed in states | Only Positive.<br>E.g.: Intelligent /\ Beautiful | Positive as well as Negative.<br>E.g.: ~Stupid /\ ~Ugly |
| 3 | 3 | Assumption for Unmentioned Literals | Closed World Assumptions:<br>Unmentioned literals are False | Open World Assumptions:<br>Unmentioned literals are Unknown |
| 4 | 4 | Equality | No Support | Equality predicate (x=y) is Built-in |
| | 5 | Types | No Support | Supported.<br>Variables can have types -> p:Person |
| | 6 | Effects | Effects are conjunctions | Conditional Effects allowed (when P:E) |
| | 7 | Effect P /\ ~Q means: | Add P & Delete Q | Add P /\ ~Q<br>Delete ~P /\ Q |
| 5 | 8 | Goals | Goals are conjunctions | Allows conjunctions & disjunctions:<br>~Poor /\ (Famous V Smart) |
| | 9 | Goals | Only ground literals | Quantified Variables |
| | 10 | Schema includes | Action name<br>Parameter list<br>Precondition<br>Effect | Action name<br>Parameter list (optional)<br>Groups of clauses labelled: Precond, Add, Delete and Update (optional) |
| 6 | 11 | Modelling actions in real world applications | Not Suitable | Suitable<br>(This inadequacy of STRIPS led to development of ADL) |

Consider an agent that delivers parcels from a warehouse to a customer within a specified time windou /chosen lime slot. a. Identify and explain the key unoertainties the agent rnay encounter during the delivery process. b. Propose two altemative delivery plans that account for the uncertainties identified above. c. List and explain the major causes of uncedainry.

### a) Key Uncertainties During Delivery Process

A parcel-delivery agent may face several uncertainties:

**1. Traffic Conditions**

- Roads may become congested unexpectedly.
- Causes delay in reaching destination.

**2. Weather Conditions**

- Rain, storms, fog, etc. may slow travel.
- Can affect road safety and speed.

**3. Customer Availability**

- Customer may not be present during delivery.
- Delivery attempt may fail.

**4. Vehicle Issues**

- Breakdown, low fuel, flat tire, etc.
- Prevents timely delivery.

**5. Incorrect / Changing Address Information**

- Customer may provide wrong address or request last-minute change.

**6. Route Blockages**

- Construction, accidents, road closures.

### b) Two Alternative Delivery Plans Accounting for Uncertainty

**Plan 1: Standard Route with Backup Route**

1. Start delivery via shortest planned route
2. Monitor traffic/weather
3. If route blocked or delayed:
   - Switch to alternate route
4. Deliver parcel within time slot

**Handles:** Traffic, road closures, weather delays

**Plan 2: Pre-Delivery Confirmation + Delivery Attempt**

1. Contact customer before dispatch
2. Confirm availability and address
3. Start delivery
4. If customer unavailable:
   - Deliver to neighbor/safe locker OR reschedule
5. Update delivery system

**Handles:** Customer absence, wrong address, failed delivery

### 3. Partial Observability / Information Uncertainty

Agent lacks complete knowledge.

Examples:

- Doesn't know if customer is home
- Doesn't know real-time road status initially

### 4. Dynamic Environment

Environment changes during execution.

Examples:

- Sudden accidents
- New delivery instructions from customer

### Final Exam-Style Summary

> A parcel delivery agent operates in an uncertain and dynamic environment where factors such as traffic, weather, customer availability, and vehicle reliability can affect successful delivery. To manage this uncertainty, the agent can employ contingency-based delivery plans such as alternate route selection and pre-delivery customer confirmation. The major causes of uncertainty include environmental factors, nondeterministic action outcomes, partial observability, and dynamic environmental changes.

map uncertainity to these.

* Observability
  - Fully (classic + condn)
    knows complete env
  - Partial (contingent)

* Deterministic
  - Classic
  - Non
    action has many outcomes
    Condn + Contigent

* Dynamicity
  - Static
  - Dynamic

Planning v/s Searching

*Types of Learning from Notes